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CRITICAL REVIEW

Integration of blockchain and life cycle assessment: a systematic literature review

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Abstract

Purpose Blockchain has the potential to bring numerous benefits to life cycle assessment (LCA), such as traceability, transparency, and accurate inventory data and assessment results. Consequently, there has been a gradual emergence of research focusing on the integration of blockchain and LCA in recent years. The aim of this study is to provide an up-to-date and comprehensive state of research on blockchain-LCA integration.

Methods This study undertook a systematic review of published articles on blockchain-LCA integration, analyzing the papers indexed so far in the Web of Science and SCOPUS. Thirty-one articles were identified, on which a bibliometric analysis was carried out. Furthermore, this study extracted and synthesized the data for the themes of benefits, barriers, frameworks and models, and case studies, and conducted detailed analysis for each theme.

Results and discussion The results indicated that the current state of research in this field is still in its early stages. Most articles proposed various potential benefits across different stages of the integration. However, the integration faced various technical, organizational, and system-related barriers. Only a limited number of articles presented frameworks, models, and case studies. Currently, there is a significant lack of case studies rooted in real-world data. Based on the results of the review, this study offered relevant suggestions for future development of the blockchain-LCA integration. Additionally, this study proposed a novel and practical generic framework to provide guidance for the continued advancement in this field.

Conclusion This work reveals that the number of studies of blockchain-LCA integration is increasing, but this field is still in the early stages. Many potential benefits and barriers to the integration were proposed, but most of them lack validation within the existing research. In particular, there is a great need for real-world case studies.

Keywords Life cycle assessment (LCA) · Life cycle inventory · Industrial ecology · Blockchain · Distributed ledger technology · Systematic review

1 Introduction

Life cycle assessment (LCA) has been one of the most accepted and used tools for the estimation of potential environmental impacts for products and services around the world (Chen et al. 2014; Guinée et al. 2011; van der Giesen et al. 2020). It has been widely used in many fields, such as waste management, assessment and improvement of specific product systems, sustainable consumption and production, evaluation of new technologies, and promotion of proactive actions (Fröhling & Hiete 2020; Hellweg & Canals 2014).

The reliability of LCA results depends on the completeness and accuracy of the input data. The life cycle inventory (LCI) analysis collects and compiles the input data for the studied product systems, which is the basis for the impact assessment of LCA (Bjørn et al. 2018). In principle, LCA should include the impacts of all processes over the entire lifecycle, which requires a large amount of inventory data (Henriksen et al. 2021). Ideally, an LCA study should consider all relevant input data, collect the specific primary data, and fulfill the characteristics of completeness, consistency, accuracy, and transparency (Bjørn et al. 2018). However, in practice most studies fail to do so due to lots of reasons: (1) data collection for LCA is time-consuming and costly, and the demand for resources

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is too great; (2) manufacturers refuse to share information due to confidentiality; (3) the data that can be obtained is often delayed because it is usually accounted based on past operating conditions rather than current conditions; (4) the life cycle value chains are possibly very complex; (5) the geographical distribution of the supply chain is too extensive (Genovese et al. 2017; Meyer et al. 2021; Modahl et al. 2013). Consequently, LCA studies usually use investigated specific data for foreground systems, and generic datasets, aggregated data, and data from literature for background data, which are universal to a certain extent and could be used for a wide range of products (An et al. 2021; Ingrao et al. 2021). However, such background data are sometimes not fully representative of locally specific and time-specific systems, and the lack of enough transparency leads to uncertainty and reliability concerns (Reap et al. 2008; Shojaei et al. 2020). Some approaches have been used to develop a variety of LCI models in order to fill the LCI data gaps, but certain challenges remain. For example, the data-driven approach faces the challenge of assessing the data quality uncertainty associated with the resulting inventory data, and the mechanistic approach requires strong discipline-specific knowledge (Zargar et al. 2022). Besides, as an iterative method, LCA requires data transparency and traceability to enhance the credibility of the results and enable it to continuously integrate the latest data to provide more reliable information to support decision-making (Björklund 2002). Due to the lack of a comprehensive system for the transparent collection and storage of LCI data, LCA currently lacks sufficient reliability for further application in the industry (Shojaei et al. 2020).

Blockchain technology can provide an excellent solution to such challenges to LCA. Blockchain is a kind of distributed ledger technology, in which all the digital events take place in a peer-to-peer network (Esmaeilian et al. 2020). Its main features include a decentralized database, information transparency, information invariance, data security, and enabling the sharing of trusted data analytics between multiple entities (Ølnes et al. 2017; Tang et al. 2019). At present, blockchain has been applied in many fields, such as financial services, insurance, healthcare, supply chain management, circular economy, and governance (Mettler 2016; Ølnes et al. 2017; Queiroz et al. 2020; Upadhyay et al. 2021). The dramatic increase in the use of RFID, sensors, and actuators and their promotion of automation reveal the potential of the integration of blockchain and the Internet of Things (IoT) to record, transmit, and process large amounts of data (Christidis & Devetsikiotis 2016; Zhang et al. 2019). For LCA, blockchain can bring a lot of potential benefits, such as reducing the time of LCA data collection, providing accurate and real-time inventory data, providing data transparency and traceability, and improving the credibility of LCA results (Asif & Gill 2022; Teh et al. 2020).

Given such potential benefits, some works in recent years have approached integrating blockchain and LCA. Though this integration can enrich knowledge and advance the development of LCA, publications in this field only increased after 2018. Some studies analyzed the potential benefits and barriers of integrating blockchain with LCA (Farooque et al. 2020; Sica et al. 2022), while some research provided a comprehensive integration framework (Zhang et al. 2020), or conducted a case study (Lin et al. 2021). Karaszewski et al. (2021) summarized the trends of blockchain applying to LCA. However, there is no systematic literature review on this topic yet. Thus, a comprehensive overview of the current status, a common understanding of the topic, and research directions for the application of blockchain technology in LCA are missing.

This study aims to fill the described gap. In our systematic review, we compile all papers related to the integration of blockchain and LCA, with the objective of screening the research types and characteristics of the current papers, providing an up-to-date and comprehensive state of research on blockchain-LCA integration. In doing so, we address the following research questions:

- How has the scientific literature on the integration of blockchain and LCA evolved in recent years, and what is the current research status?
- What themes are the current studies mainly focusing on? Which areas are covered well, and which aspects are missing? What are the potentials and limits of blockchain for LCA?
- How can an integration of blockchain and LCA be achieved? What are the directions for future development?

This paper proceeds as follows: Sect. 2 presents methods for this systematic review; Sect. 3 shows the results, including descriptive analysis and thematic analysis; Sect. 4 discusses the current state of research on blockchain-LCA integration, provides suggestions for the future development, and proposes a new framework for the integration of blockchain and LCA in the future; and Sect. 5 presents the final conclusion.

2 Methods

Referring to the PRISMA systematic review protocol (Moher et al. 2015), we conducted the systematic review of the current state of integration of blockchain and LCA in four steps: locating studies, study selection and evaluation, analysis and synthesis, and reporting and using the results.

2.1 Locating studies

Web of Science is the oldest, most widely used, and authoritative database of research publications and citations in the world (Birkle et al. 2020), while SCOPUS is one of the largest curated abstract and citation databases, covering global and regional scientific publications with selected high-quality data (Baas et al. 2020). They have been the two most widely used databases for bibliometric analysis (Singh et al. 2021). Therefore, we chose Web of Science and SCOPUS to search for publications about blockchain and LCA. Two groups of expressions were used for the search: one group was for blockchain, and the other was for LCA. The first group consisted of the expressions of blockchain, block chain, and distributed ledger technology, while the second group included the expressions of life cycle assessment, life cycle analysis, and LCA. For Web of Science, the search string was (TS = (blockchain) OR TS = (block chain) OR TS = (distributed ledger technology)) AND (TS = (life cycle assessment) OR TS = (LCA) OR TS = (life cycle analysis)). For SCOPUS, the search string was (TITLE-ABS-KEY(blockchain) OR TITLE-ABS-KEY(block chain) OR TITLE-ABS-KEY(distributed ledger technology)) AND (TITLE-ABS-KEY(life cycle assessment) OR TITLE-ABS-KEY(life cycle analysis) OR TITLE-ABS-KEY(LCA)).

2.2 Study selection and evaluation

Figure 1 shows the detailed steps of study selection and evaluation. We searched in June 2024 and found 210 publications in Web of Science and 456 publications in SCOPUS and managed them with EndNote. After removing the duplicates, we reached 522 publications. Then we selected papers based on the inclusion and exclusion criteria: (1) peer-review journal papers were included while grey publications were excluded except conference papers since the researchers from the IT field are more inclined to publish conference papers than journal papers; (2) papers related to blockchain for LCA and blockchain-LCA integration were included, while the papers about LCA of blockchain were excluded; and (3) only papers in English were included. After removing some grey publications and scanning titles, abstracts, and keywords, the set for further investigations comprised 109 papers. Next, we read the full texts of the 109 papers to conduct the content-related check and excluded those papers not related to the integration of blockchain and LCA, yielding 27 papers. Finally, we completed the snowball process for the included papers (Wohlin 2014) and eventually found four related papers. As a result, a total of 31 relevant papers were identified.

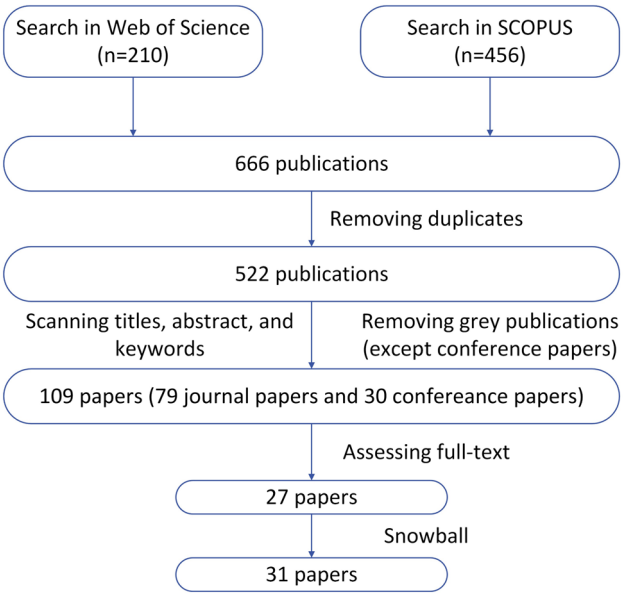


Fig. 1 Flow diagram of the study selection and evaluation process

2.3 Analysis and synthesis

We performed the analysis and synthesis as shown in Fig. 2. For a descriptive overview, the selected papers were examined concerning the temporal, geographical, journal, and conference distribution. Besides, the industrial sectors involved in the studies about the integration of blockchain and LCA were also identified. We conducted qualitative data extraction for all the selected papers. We extracted and synthesized the data for the themes of benefits, barriers, frameworks and models, and case studies. We also conducted a detailed analysis of each theme. Finally, we provided our perspectives on the current research state and suggestions for future development, proposed a novel framework to provide guidance for continued advancement, and presented the implications and limitations of this study.

3 Results

3.1 Descriptive analysis

The descriptive analysis, in general, is characterized by the comparable small number of papers integrating blockchain and LCA, as well as their citations. As shown in Fig. 3, the first paper on the integration of blockchain and LCA was published in 2018 due to the short history of blockchain. Three papers were published in 2018, while only one was published in 2019. After that, the number of papers on this topic increased, with eight papers in this field published in 2022. In 2023, only six papers were published, and it is not

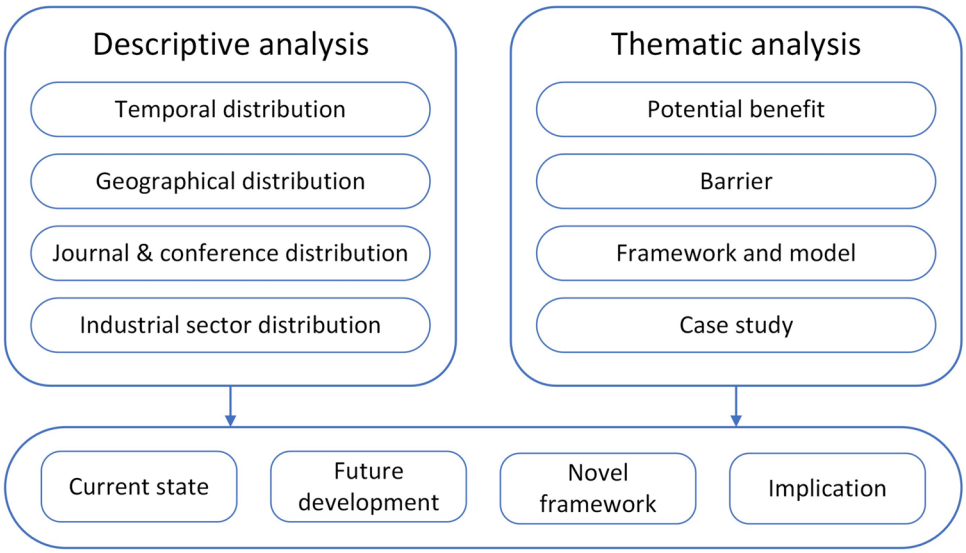


Fig. 2 Analytical framework

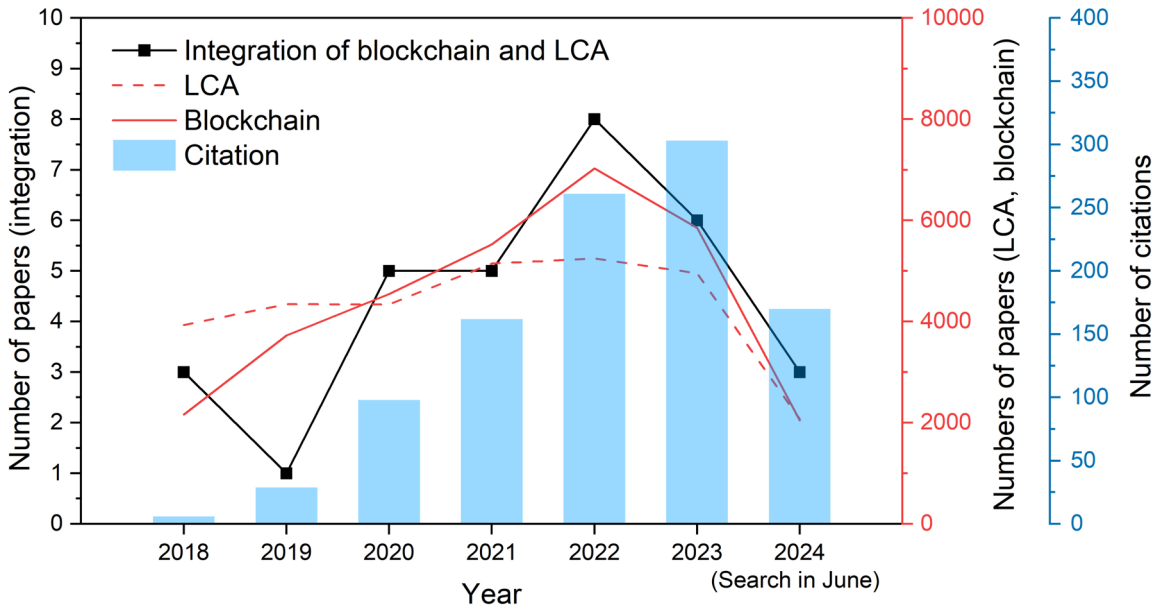


Fig. 3 Temporal distribution of selected papers about the integration of blockchain and LCA, papers on the topic of LCA, papers on the topic of blockchain, and the citations of the papers about the integration of blockchain and LCA

clear if this is indicating a decline in attention to this topic. As we searched in June 2024, we only found three papers published in 2024. In order to provide a clear picture of the publication trend about the integration of blockchain and LCA, we compared it to the publication trends of papers on LCA and papers on blockchain. The growth rate of papers about the integration of blockchain and LCA is higher than those of these two topics. Regarding the citations, all articles have accumulated a total of 1029 citations to date, showing a consistent upward trend excluding 2024, with a peak at 303 citations in 2023.

As the countries that published the most LCA papers (10,161) and the most blockchain papers (11,866), respectively, USA and China published the most papers about the integration of blockchain and LCA, reaching four, followed by Germany (three), Australia (three), Italy (three), the UK (two), Canada (two), and New Zealand (two) (Fig. 4). According to the country classifications by income by the World Bank (The World Bank 2023), high-income economies published 23 papers in this field, accounting for 74%, while upper-middle-income

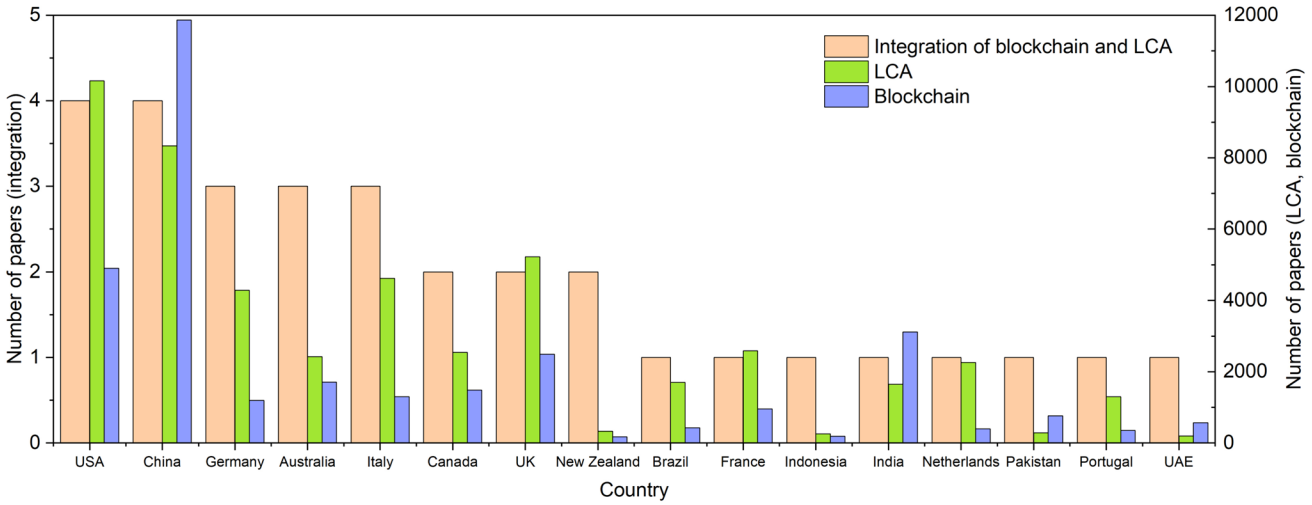


Fig. 4 Geographical distribution of selected papers about the integration of blockchain and LCA, papers on the topic of LCA, and papers on the topic of blockchain

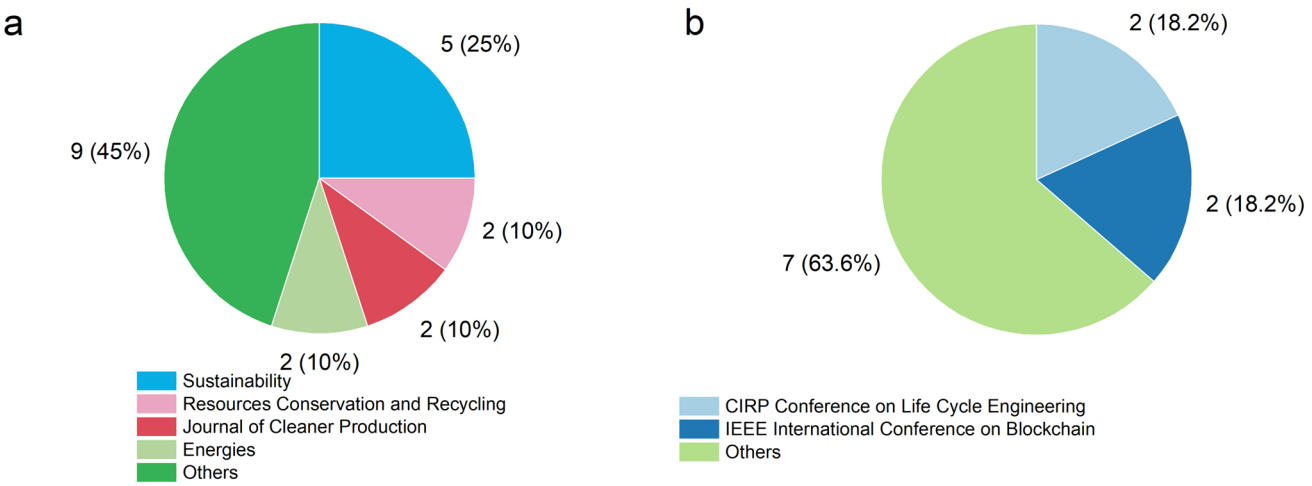


Fig. 5 Distribution of the papers among journals (**a**) and conferences (**b**)

economies and lower-middle-income economies published six and two articles prospectively.

Figure 5 shows the paper distribution in journals and conferences. The journal *Sustainability*[1] accounts for the largest number of papers with five, while *Resources, Conservation and Recycling*, *Journal of Cleaner Production*, and *Energies* host two papers each. Other journals, such as *The International Journal of Life Cycle Assessment*, *Applied Sciences*, *Built Environment Project and Asset Management*, *Corporate Social Responsibility and Environmental Management*, *Environment Systems and Decisions*, *Building and Environment*, *Egyptian Informatics Journal*, *Science of The Total Environment*, and *Computers & Industrial Engineering*, contribute one paper, respectively. As for the conferences, the CIRP Conference on Life Cycle Engineering and IEEE International Conference on Blockchain host two papers each, while the International Conference on Environment Science and Engineering, International Conference on Enterprise Information Systems, International Conference of Architectural Science Association, IEEE International Conference on Computer and Communications, IEEE Vehicle

[1] For the journal *Sustainability* and its publishing house, it is more and more discussed whether they should be categorized as predatory (Oviedo-Garcia, 2021). However, since the classification of the journal and the publishing house does not necessarily say something about the individual article, and *Sustainability* is still indexed in the used sources Web of Science and Scopus, we decided to include the articles found through the described systematic literature review process.

Power and Propulsion Conference, Communications in Computer and Information Science, and IFIP International Conference on Advances in Production Management Systems contribute one paper, respectively.

We classify the papers according to the industrial fields they focused on, as shown in Fig. 6. Papers involving more than one industrial field or not focusing on specific industries accounted for the majority, reaching 14 papers. Six papers focused on the construction sector, three on the food manufacture, three on the textile manufacture, and two on the transport equipment. There was one paper each about leather manufacture, electronic products, and fishing and aquaculture.

3.2 Thematic analysis

In order to answer the research questions, we analyzed the articles according to their thematic focus on the integration of blockchain and LCA. Specifically, four themes were identified: the potential benefits, the barriers, the frameworks and models, and case studies.

3.2.1 Potential benefit

There are 18 papers covering the potential benefits of the integration of blockchain and LCA. We divided these benefits into six groups according to the different stages of LCA: goal and scope, inventory analysis, impact assessment, interpretation, all the phases, and the application. The findings from the literature are explained in the following paragraphs, which are summarized in Table 1.

For the goal and scope, three papers report the potential benefits of the integration of blockchain and LCA. As the blockchain system can provide large amounts of information about the whole lifecycle of the products and services as well as the supply chains, it can provide more objective insights and reduce the need for subjective judgment, which helps to choose a more reasonable functional unit (Carrieres et al. 2021; Sica et al. 2022; Zhang et al. 2020). Besides, for conducting an LCA, the scope of it is usually constrained by data availability, limited time for data collection, etc. The application of blockchain technology can improve the traceability and transparency of the supply chain to remove such data acquisition constraints, making it easier to define the goal and scope of LCA studies (Zhang et al. 2020).

All 18 papers mentioned the potential benefits for the inventory analysis stage. The data collection of LCA is very time-consuming and cost-intensive, while the integration of blockchain and LCA can effectively reduce the time and cost required and automatically process relevant data (Figueiredo et al. 2022; Rolinck et al. 2021). Besides, it helps to improve data quality; provide more accurate, realistic, and real-time information; and reduce uncertainty (Asif et al. 2022; Karaszewski et al. 2021; Lin et al. 2021). Moreover, data integrity can be ensured to account for all possible inputs in complex supply chains (Carrieres et al. 2021; Zhang et al. 2020). This integration can ensure instant traceability and transparency of product information and data sources (Asif & Gill 2022; Kouhizadeh & Sarkis 2018). For the long-term development of LCA, it can also provide new standards for LCA data and help to establish a new transparent database (Teh et al. 2020). Furthermore, it can serve in the inventory analysis of specific industries, such as estimating energy consumption of transportation with the integration of GPS and assessing material regeneration and restoration effects in circular supply chains (Li & Wang 2021; Zhang et al. 2020).

Seven papers provide the potential benefits for the impact assessment stage. With IoT, it can automatically calculate the environmental impacts and continuously record a large number of results under different periods and working conditions to form big data to explore valuable and true insights (Asif & Gill 2022; Karaszewski et al. 2021; Teh et al. 2020). Besides, it has the potential to provide a more fine-grained, accurate, and relevant impact assessment due to the inventories (Carrieres et al. 2021; Zhang et al. 2020). In addition, applying it to a broader supply network can help to identify

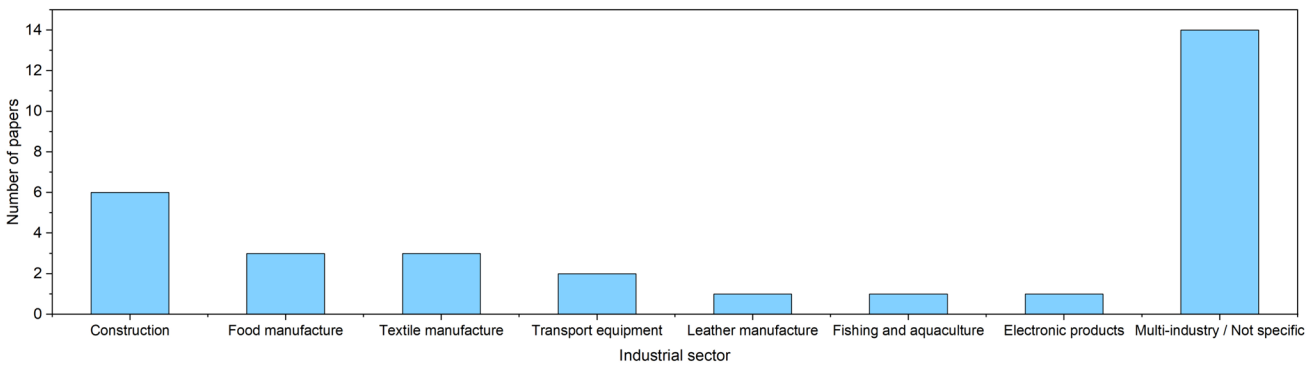


Fig. 6 Number of papers by classification of involved industrial sectors

Table 1 Potential benefits of integrating blockchain and LCA

LCA phase	Potential benefit	Source
Goal and scope	1. Give an objective insight into the functional unit 2. Make it easy to define the goal and scope because of the traceability and transparency provided	1. Carrieres et al. (2021), Sica et al. (2022), Zhang et al. (2020) 2. Zhang et al. (2020)
Inventory analysis	1. Provide accurate, actual, and real-time information about products and services rather than estimation, improve data quality, and reduce information uncertainty in an LCA analysis 2. Decrease the costs and time required for data collection 3. Ensure instant traceability and transparency of product data and data sources, especially when integrating blockchain with IoT 4. Ensure maintained data integrity 5. Provide a new information standard for LCA data 6. The integration with blockchain and GPS can help to automatically estimate the energy consumption of transport 7. Help to build a free and transparent LCA database 8. Enable the accounting of all possible inputs of complex supply chains 9. Make it possible to quantify the material regeneration and restoration effects within a circular supply chain 10. Allow the automated documentation of LCA-relevant data and automated processing of data	1. Alwi et al. (2024), Asif et al. (2022), Figueiredo et al. (2022), Kouhizadeh and Sarkis (2018), Li and Wang (2021), Lin et al. (2021), Olanrewaju et al. (2022), Sica et al. (2022), Soufi et al. (2023), Teh et al. (2020), Yang et al. (2023), Zhang et al. (2020) 2. Carrieres et al. (2021), Figueiredo et al. (2022), Kouhizadeh and Sarkis (2018), Sica et al. (2022), Teh et al. (2020), Yang et al. (2023) 3. Alwi et al. (2024), Asif and Gill (2022), Asif et al. (2022), Chen et al. (2024), Figueiredo et al. (2022), Kouhizadeh and Sarkis (2018), Soufi et al. (2023), Olanrewaju et al. (2022), Piron et al. (2024), Rolinck et al. (2021), Teh et al. (2020), Zhang et al. (2020) 4. Figueiredo et al. (2022), Zhang et al. (2020) 5. Kouhizadeh and Sarkis (2018) 6. Li and Wang (2021) 7. Teh et al. (2020) 8. Carrieres et al. (2021), Zhang et al. (2020) 9. Zhang et al. (2020) 10. Carrieres et al. (2021), Rolinck et al. (2021)
Impact assessment	1. Integrate blockchain and IoT to continuously record data transactions to form big data, and explore useful and true insights with big data analytics 2. Provide a more fine-grained, accurate, and relevant impact assessment 3. Identify environmental impacts across global supply chains 4. Automate the environmental impact calculation	1. Asif and Gill (2022), Karaszewski et al. (2021), Zhang et al. (2020) 2. Carrieres et al. (2021), Sica et al. (2022), Teh et al. (2020), Zhang et al. (2020) 3. Teh et al. (2020) 4. Carrieres et al. (2021), Rolinck et al. (2021), Sica et al. (2022), Yang et al. (2023)
Interpretation	1. Improve the accuracy and credibility of LCA results 2. Build consumer trust with the possibility of communicating the real-time impacts of products 3. Enable data analysis from different perspectives and produce graphical insights of LCA data	1. Asif et al. (2022) 2. Carrieres et al. (2021), Zhang et al. (2020) 3. Zhang et al. (2020)
All phases	1. Store environmental information of a product through its life cycle to better manage its recycling and disposal 2. Help to facilitate the achievement of SDG9, 12, and 13 3. Keep data security and privacy, promise that enciphered private information can only be read or edited by the user who has the key 4. Make it easy to prevent cheating 5. Meet the information requirements and decision-making process of a contemporary business by sharing immutable and irreversible information in real-time	1. Kouhizadeh and Sarkis (2018) 2. Sica et al. (2022), Teh et al. (2020) 3. Alwi et al. (2024), Chen et al. (2024), Lin et al. (2021), Yang et al. (2023) 4. Alwi et al. (2024), Lin et al. (2021) 5. Teh et al. (2020)

Table 1 (continued)

LCA phase	Potential benefit	Source
Application	1. Reduce information uncertainty and provide better modeling inputs and outputs for eco-design 2. Verify whether standards are being implemented upstream and downstream in the supply chain regarding various sustainability-related factors 3. Products and inventories can be tagged with related environmental information 4. Aid businesses to improve the monitoring and management of the environmental sustainability performance of their supply chain activities 5. Help firms achieve operational excellence and improve competitiveness by building superior customer trust and loyalty with data transparency 6. Help governments monitor environmental impacts 7. Make environmental accounting easier 8. Inform better logistics and supply chain design decisions for improving supply chain sustainability performance	1. Carrieres et al. (2021), Kouhizadeh and Sarkis (2018), Zhang et al. (2020) 2. Teh et al. (2020) 3. Teh et al. (2020) 4. Zhang et al. (2020) 5. Zhang et al. (2020) 6. Zhang et al. (2020) 7. Zhang et al. (2020) 8. Zhang et al. (2020)

environmental impacts across global supply chains (Rolinck et al. 2021; Sica et al. 2022).

The integration of blockchain and LCA can also bring benefits to the interpretation stage. It improves the accuracy and credibility of LCA results and builds consumer trust with the possibility of communicating the real-time impacts of products (Asif et al. 2022; Carrieres et al. 2021). In addition, it has the potential to contribute to data analysis from different perspectives and provide graphical insights into LCA data (Zhang et al. 2020). However, there is a lack of research presenting the benefits of the integration for critical reviews of LCA studies. From our perspective, it makes it easier for the reviewer to validate the data quality as the blockchain can provide transparent and trackable data for the life cycle inventory analysis. Besides, it can reveal involved industries along the supply chains more transparently, which helps the organizer to find experts from related fields more appropriately for the review panel.

The integration has some features that can benefit all phases of LCA. It can make it easy to prevent cheating, keep data security and privacy, and promise that enciphered private information can only be read or edited by the user who has the key (Alwi et al. 2024; Lin et al. 2021). It also stores environmental information for the entire life cycle of products, helping in the management as well as recycling and disposal (Kouhizadeh & Sarkis 2018). By sharing immutable and irreversible information at all stages of LCA in real-time, it can provide references for enterprise decision-making (Teh et al. 2020). Moreover, it helps LCA better promote the achievements of SDGs 9, 12, and 13 (Sica et al. 2022).

It has great application potential and benefits for various fields. First, the integration makes environmental accounting easier for all stakeholders (Zhang et al. 2020). In the field of ecological design, it can provide better model input and input with a decrease in information uncertainty (Carrieres et al. 2021; Kouhizadeh & Sarkis 2018). As for the supply chain, it helps standardize product environmental information and verify the implementation of sustainable standards for the upstream and downstream of the supply chain (Teh et al. 2020). Besides, it can provide a reference for logistics and supply chain design decisions and help companies monitor and manage the environmental situations of the supply chains, thereby improving sustainability performance (Zhang et al. 2020). Enterprises can also use it to increase data transparency to improve customer trust and loyalty (Zhang et al. 2020). For the government, the integration provides transparent environmental information, making it easier for the government to monitor environmental impacts (Zhang et al. 2020).

3.2.2 Barrier

Seven papers provided information on the barriers to the integration of blockchain and LCA. Carrieres et al. (2021) compared and systematically summarized the barriers of blockchain-based LCA from various aspects. Farooque et al. (2020) prioritized the important obstacles to the integration through the Fuzzy DEMATEL analysis and highlighted the obstacles in technology immaturity, challenges in collecting real-time data, and lack of policies/support. As shown in Table 2, the barriers can be divided into five groups: technological, organizational, inter-organizational, external, and system-related barriers (Carrieres et al. 2021; Farooque et al. 2020).

The number of technical barriers is the highest, reaching eight. The immaturity of relevant technologies is one of the main barriers (Farooque et al. 2020). The current blockchain system makes it difficult to cope with the increasing number of transactions and handle a large number of simultaneous

Table 2 Barriers to integrating blockchain and LCA

Category	Barrier	Source
Technological	1. Immaturity of technology	1. Farooque et al. (2020)
	2. Difficulty of blockchain in dealing with a growing number of transactions	2. Carrieres et al. (2021)
	3. Difficulty to deal with the large number of simultaneous transactions	3. Carrieres et al. (2021)
	4. Limitations in data transmission rate and data storage	4. Rolinck et al. (2021), Zhang et al. (2020)
	5. Challenges for collecting supply chain data in real-time	5. Farooque et al. (2020)
	6. It can be difficult to operate with other systems	6. Carrieres et al. (2021)
	7. Limitation of physical parts: the available energy sources, the need to supply extended hardware computing power, the network transfer infrastructure	7. Smetana et al. (2018)
	8. Challenges about prevention of data manipulation prior to uploading data to the blockchain network	8. Rolinck et al. (2021), Zhang et al. (2020)
Organizational	1. Cost and time for setup and maintenance	1. Carrieres et al. (2021), Sica et al. (2022)
	2. Lack of management commitment and support	2. Carrieres et al. (2021), Farooque et al. (2020)
	3. Lack of technical expertise and knowledge	3. Carrieres et al. (2021)
	4. Lack of standards or appropriate methods, tools, or metrics available for blockchain implementation	4. Carrieres et al. (2021), Rolinck et al. (2021)
	5. Hesitation and opposition to altering and replacing already existing systems	5. Carrieres et al. (2021), Farooque et al. (2020)
	6. Lack of new organizational policies for using technology	6. Farooque et al. (2020)
Inter-organizational	1. Challenges of information disclosure and data sharing by stakeholders	1. Carrieres et al. (2021), Farooque et al. (2020), Sica et al. (2022), Teh et al. (2020)
	2. Problem of collaboration, communication, and coordination in the supply chain	2. Carrieres et al. (2021), Farooque et al. (2020)
	3. Different organizational cultures can hinder the adoption of blockchain	3. Carrieres et al. (2021), Sica et al. (2022)
	4. Challenges in integrating sustainable practices and Blockchain technology through supply chain management	4. Farooque et al. (2020)
External	1. Lack of governmental regulation	1. Carrieres et al. (2021), Farooque et al. (2020), Rolinck et al. (2021)
	2. Lack of external stakeholders' involvement and support	2. Carrieres et al. (2021), Farooque et al. (2020)
	3. Lack of rewards and encouragement programs	3. Farooque et al. (2020)
System-related	1. Doubt about Blockchain technology due to negative public perception	1. Carrieres et al. (2021), Farooque et al. (2020)
	2. Security challenge	2. Farooque et al. (2020), Sica et al. (2022)
	3. The allocation of decision management rights	3. Teh et al. (2020)

transactions (Carrieres et al. 2021). The system data transmission rate and data storage also have certain limitations (Rolinck et al. 2021; Zhang et al. 2020). Besides, there are also many challenges in collecting supply chain data in real-time (Farooque et al. 2020). As for compatibility, the blockchain-LCA system currently has some obstacles in operating with other systems. There are also some physical constraints, including the energy sources, the extended hardware computing power, and the network transfer infrastructure (Smetana et al. 2018). Although the blockchain performs well in preventing cheating, there are still challenges in preventing data manipulation before uploading data to the blockchain network (Rolinck et al. 2021; Zhang et al. 2020).

There are six organizational barriers to the integration of blockchain and LCA. The system maintenance and setup require a lot of cost and time (Sica et al. 2022). Besides, the application of the system is lacking in various aspects concerning the organization: management commitment and support, technical expertise and knowledge, standards or appropriate methods, tools or metrics available for blockchain implementation, and new organizational policies for using blockchain (Carrieres et al. 2021; Rolinck et al. 2021). Another challenge is the hesitation and opposition about altering and replacing already existing systems in the management of companies (Farooque et al. 2020). As for the inter-organizational barriers, there are still many problems with collaboration, communication, and coordination in the supply chains (Farooque et al. 2020). Various stakeholders still have concerns about information disclosure and data sharing (Teh et al. 2020). The different organizational cultures also hinder the adoption of blockchain (Carrieres et al. 2021; Sica et al. 2022). Moreover, many challenges remain to integrate sustainable practices and blockchain technology through supply chain management (Farooque et al. 2020).

There are some external obstacles to the application of the blockchain-LCA integration system, such as the lack of governmental regulation, external stakeholders' involvement and support, and rewards and encouragement programs (Carrieres et al. 2021; Farooque et al. 2020; Rolinck et al. 2021). At the same time, there are some system-related barriers. Some members of the public have negative perceptions of blockchain (Carrieres et al. 2021). Besides, the integration system also has some security challenges (Farooque et al. 2020; Sica et al. 2022). And there are different views on the allocation of management rights (Teh et al. 2020).

3.2.3 Framework and model

Two papers aimed to develop general frameworks for the integration of LCA and blockchain. Zhang et al. (2020) put forward a generic system architecture for the integration of blockchain and LCA, including four layers: the infrastructure layer, the blockchain service layer, the application layer, and the user layer. The basic layer of the system is composed of the operating system and hardware, combined with IoT, using various sensors for data collection, transmission, and recording throughout the life cycle stages. Besides, blockchain is used in the service layer for data processing, especially inventory analysis and impact assessment results. It also enables data traceability and hotspot analysis and is integrated with big data to provide useful and true insights. The application layer covers six applications: energy-saving management, ecosystem quality management, waste management, reuse/recycling management, product information management, and uncertainty treatment management. Finally, the user layer includes governments, NGOs, consumers, and enterprises. Taranto et al. (2023) proposed an energy-based sustainability framework for global and scalable payment systems, which integrates LCA to assess environmental impacts, internalize external costs, and utilize blockchain technology for transactions and record-keeping. The combination of LCA and blockchain for the payment tracking systems enhances transparency, increases the resolution of environmental accounting measures, and facilitates broader cross-industry collaboration.

Several studies proposed blockchain-LCA frameworks for specific industrial sectors. Shou and Domenech (2022) conducted the iterative Delphi-inspired method, built the blockchain-LCA integration framework blueprint for circular fashion with semi-structured interviews with experts, and refined it through an iterative feedback process via email. The framework is organized based on the four stages of LCA and explains how the blockchain serves each stage of LCA for the textile industry. After defining the scope, stakeholders along the value chain are registered on the blockchain platform. During the inventory analysis stage, data can be collected and verified using the IoT and optical character recognition (OCR) tools. After that, it uses the inventory generated by blockchain to calculate the environmental impacts, and then effectively deliver the LCA results to users with visualization tools. Referring to the Emission Trading Scheme, Fu et al. (2018) combined the Emission Trading Scheme and developed a blockchain-based life cycle carbon emission accounting framework specifically tailored for the textile production industry. It involves four key entities: the authority, the auditor, the firms, and the individuals. The authority serves as the initiator and regulator, requiring firms to participate in the system and designating auditors to assess carbon emissions based on data provided by the blockchain and generate reports. This framework also discloses environmental information to support individuals' consumption decisions and strengthens the connection between individuals and companies. For aircraft maintenance, repair, and overhaul, Rolinck et al. (2021) developed a conceptual framework for blockchain data management for LCA, which automates the collection and recording of data and the calculation of LCA results. In that framework, each organization about aircraft collects data in its own information system and chooses to upload the original data or the further processed data through blockchain for the calculation of LCA. It avoids disclosing important information while enabling effective calculation and sharing of LCA results. Blumberg and Sibilla (2023) developed a model for a carbon accounting and trading platform (CATP) specifically focused on carbon taxation and carbon trading in the United Kingdom's construction sector. This model employs LCA to track the carbon emissions throughout the entire lifecycle of buildings and uses such information as a basis for levying carbon taxes with blockchain technology.

Seven studies suggested blockchain frameworks for LCA from technical perspectives. Shojaei et al. (2020) developed a blockchain network framework for the construction sector, which could be integrated with Building Information Modeling (BIM) and serve as the infrastructure for LCA. Three types of participants were defined, including manufacturer, trader, and regulator, which could trade via the network based on the certificates issued by regulators. Hyperledger fabric was used to develop a consortium blockchain, which gave the network higher efficiency and scalability than the public ones. Kumar et al. (2023) focused on 3D printing and proposed a lifecycle tracking and impact assessment framework based on Ethereum. This framework assigns a unique batch identity number to every process and material in the supply chain, including different batches and qualities of raw materials, intermediate products, final products, associated companies, equipment, as well as the waste and carbon emissions generated. All of the information is authenticated by third parties. Hao (2022) focused on the chips and developed a blockchain-based carbon footprint tracking system framework using the Hyperledger Fabric

platform with the Go programming language. This framework involves stakeholders from various stages of the chip supply chain, who collect carbon emission data related to material, processing, transportation, and sales processes and upload them to the Super Ledger Fabric, creating a blockchain system for tracking carbon footprints within the supply chain. Lin et al. (2021) designed a meticulous technical blockchain-LCA network and explained in detail how information is transferred and stored with blockchain technology. It defined three types of nodes in the network for information transfer, including sending nodes, receiving nodes, and transfer nodes, and elaborated on the entire workflow through the nodes in the blockchain network. Panza et al. (2023) suggested a framework for the digital product passport (DPP) to assess and record the absolute environmental sustainability with LCA and the planetary boundaries. It details the various types of data that should be collected and the Cyber-Physical Systems for its implementation. da Cruz et al. (2020) presented a model for a smart contract-based platform built on Ethereum, which is designed to record monthly product production and supply chain information, aiming to track and calculate the carbon footprint of both products and organizations. It also allows consumers to track the supply chain and access carbon footprint information through a mobile app. Alwi et al. (2024) developed a technological framework combining blockchain and big data for the fishery supply chain, which integrates life cycle carbon footprint analysis. This framework is used to efficiently trace the real information in the supply chain, facilitating a comprehensive evaluation based on product quality and sustainability. Additionally, the framework was tested through simulation, white-box evaluation, and complexity analysis, demonstrating its practical applicability for enhancing the fishery supply chain.

3.2.4 Case study

There are four papers containing case studies about the integration of blockchain and LCA. Lin et al. (2021) developed a Python-based blockchain-LCA model for a virtual case simulation. It used the simplified data from the organic chemical supply chain to demonstrate the application of the model. The simulation results showed that the application of the network could improve the accuracy of the environmental impacts and revealed the periodic fluctuation of carbon emissions of the supply chain. Fu et al. (2018) focused on the proposed blockchain-based life cycle carbon emission accounting framework. They developed a simplified clothing manufacturing supply chain to perform case analysis using virtual data for system validation. Additionally, they introduced formulas for the determination of the uncertainties. Shakhbulatov et al. (2019) applied GPS with blockchain to collect and transfer data for food transportation and calculated LCA results with simulated data. They developed the carbon footprint chain (CFC) system with a blockchain consensus algorithm, which showed higher performance in blockchain creation time and average transaction time compared to the Bitcoin blockchain. Carrieres et al. (2022) conducted an LCA case for worsted wools based on the data collected from the blockchain platform and compared it with the LCA case based on traditional data sources. It showed that the impacts calculated based on the specific data collected with the blockchain network changed drastically from those calculated with generic data. The study suggested that blockchain traceability data could provide specific and differentiated data for a more representative assessment of LCA. For these four case studies, three used the simulation data instead of real-world data to calculate the results, while the other one used the real-world data collected with blockchain but did not integrate the blockchain with LCA into one system. There is still a lack of a representative application case with real-world data for the integration of blockchain and LCA.

4 Discussion

4.1 Current state of research

The descriptive analysis gives us a clear picture of the current research status of the integration of blockchain and LCA. Based on the temporal distribution of papers and citations and the journal and conference distribution of papers, we can find that the research is still in its infancy, with the first article published in 2018, and only 31 articles have been published, while most of them were published in journals and conferences with limited impact. Furthermore, the thematic analysis shows that a majority of these articles primarily focused on the analysis of potential benefits and barriers without verification in practice. At the same time, only a handful of them delved into the development of frameworks, models, and case studies. There remains a deficiency in practical application research. We can identify that there are still many unresolved issues in the research on blockchain-LCA integration.

There is a noticeable lack of both quantity and quality when it comes to case analyses. While many articles analyzed the potential benefits and barriers of blockchain-LCA integration, these analyses are largely based on expert interviews or the authors' personal comprehension of the field. Regrettably, most of these proposed potential benefits and barriers lack real-world case examples to substantiate their claims. This scarcity can be attributed to the early research stage, which has a limited number of published cases. Moreover, the existing case studies often failed to account for the essential benefits that blockchain-LCA integration offered.

Of the existing case studies, three employed simulated data for testing the proposed blockchain-LCA integration models, which failed to demonstrate the key benefits such as transparency, real-time and accurate data collection, and data auto-processing. The remaining case study utilized high-precision data collected through blockchain for conducting LCA, emphasizing the accuracy it brings to inventory analysis and impact assessment. However, it did not offer a true integration of blockchain with LCA, so it cannot showcase the potential benefits of automated impact calculations and real-time environmental impact tracking. At present, there is a lack of truly representative blockchain-LCA integration cases in practical applications.

Simultaneously, there is a notable absence of a standardized, technically detailed, and executable framework. As the research on blockchain-LCA integration is still in its infancy, various articles presented their interpretations, resulting in various frameworks and models without a standardized consensus. Some articles proposed general frameworks for blockchain-LCA integration, but these frameworks are often more conceptual and lack detailed execution guidance. Other articles developed industry-specific blockchain-LCA integration frameworks, which adapted the general framework to the unique circumstances of the industries and provided descriptions of operational mechanisms to enhance their applicability. Nonetheless, they often still lacked the required level of detail for implementation. Some articles attempted to create technically feasible models, but they typically had shortcomings. For instance, most models remained in the descriptive stage of technical mechanisms rather than the actual development of the technology. In cases where there is technical development, it is often limited to specific aspects of the system, such as programming for carbon footprint calculation and data uploading using blockchain, but there has been insufficient exploration of other parts of the system, assuming that they can adapt to the operation of the program. Research about some critical aspects, such as the technical intricacies of employing the IoT for data collection in a blockchain-LCA integration system, transforming data into the required LCA data format, and automating the LCA calculations, remain largely unexplored. Discussions on the interaction between blockchain and LCA tools, LCA software optimization and plugin development for blockchain, or the possibility of creating specialized LCA software are all insufficiently discussed.

Furthermore, there is a deficiency in analyzing the practical application of blockchain-LCA integration in society. Different industrial sectors may have diverse requirements for blockchain-LCA integration, necessitating industry-specific analysis. Currently, there is an absence of research examining the applicability and ability of blockchain-LCA integration to meet the diverse needs of various industrial sectors. Besides, there is a lack of discussion about the disadvantages of the integration of blockchain and LCA. Additionally, the research focused on policy analysis and business model analysis in the context of blockchain-LCA integration is still insufficient.

4.2 Future development

Considering the current status of blockchain-LCA integration and the challenges it faces, we propose the following suggestions for its future development.

There is a great need to promote research on various aspects in the field of blockchain-LCA integration. It is essential to conduct in-depth analyses of the specific requirements and support systems needed in various industrial sectors to develop blockchain-LCA integration systems tailored to their unique needs. Besides, it is necessary to propose a more comprehensive and practical general comprehensive blockchain-LCA integration framework containing more technical details to guide future development. In addition, considerable effort should be directed toward the research of case studies to thoroughly validate the proposed benefits, learn from experiences to improve multiple details of the system and explore solutions for overcoming various challenges in practice. Our following step is to use the framework we developed to conduct case studies based on real-world data to supplement the lack of existing research. As the authors are both from a research background in industrial ecology, which may cause bias in the extraction and synthesis of the results in this study, experts from diverse fields could be invited to work together in the future, containing standardizing models and guidelines. This may also include the creation of a universal platform for data sharing, standardized methods for data coordination and estimation, consistent data collection and reporting, and the establishment of standardized management approaches. Furthermore, enhancing the system with various technical details to make it more practical is of utmost importance. This could involve developing software or plugins that facilitate seamless interaction with LCA software and ensure compatibility with IoT and related hardware. In the future, there is a potential to create a new LCI database based on data collected through blockchain technology. Once a highly operational model is in place, real-world industrial data from different sectors should be utilized for case studies to validate and fully exploit the benefits of blockchain-LCA integration. In the future, the integration should be gradually expanded for economic and social assessments, leading to the development of a blockchain-LCSA integration system that further promotes sustainability. Furthermore, there is a critical need to intensify research on policy analyses and business models.

On the organizational and societal level, steps should also be taken to facilitate the development of blockchain-LCA integration. Governments should introduce a series of

policies and establish relevant programs to promote blockchain-LCA integration. Simultaneously, they should enhance the management of the system and coordinate various stakeholders on the platform, especially concerning data access management. Creating specialized forums, communities, alliances, and the inclusion of experts from related fields will be instrumental in nurturing more professionals in this domain.

4.3 Proposed framework for the integration of blockchain and LCA

In order to guide the practical implementation of blockchain-LCA integration, now we proposed a new generic blockchain-LCA integration framework, which can provide a trackable, automatic, and precise environmental impact assessment from a life cycle perspective. Building on the analyzed literature, Fig. 7 depicts such.

4.3.1 Proposed framework

The proposed framework comprises five layers: the product system layer, data collection and processing layer, blockchain platform layer, blockchain-based LCA layer, and service and application layer.

The product system layer encompasses the entire life cycle of the products, including sourcing, manufacturing and assembly, distribution, retail and usage, and end-of-life treatment (EoL). The data collection and processing layer is responsible for the initial gathering and processing of raw data. This layer uses IoT systems for data collection, incorporating sensor systems, local information systems, and data transmission networks. It begins by collecting data using various intelligent sensors, such as those for monitoring electricity, water, and gas consumption, as well as identifying materials, calculating consumption quantities, and measuring emissions. Data are then stored and transmitted via local servers and networks. Subsequently, specialized software is employed to perform initial processing on the raw data, encompassing data cleaning, format conversion, and data classification and aggregation. This facilitates further handling of the data using blockchain technology.

An existing blockchain platform is refined, or a new one is developed to meet the requirements of recording, transmitting, and processing the data for integration. To enhance the traceability of supply chains, the platform features six

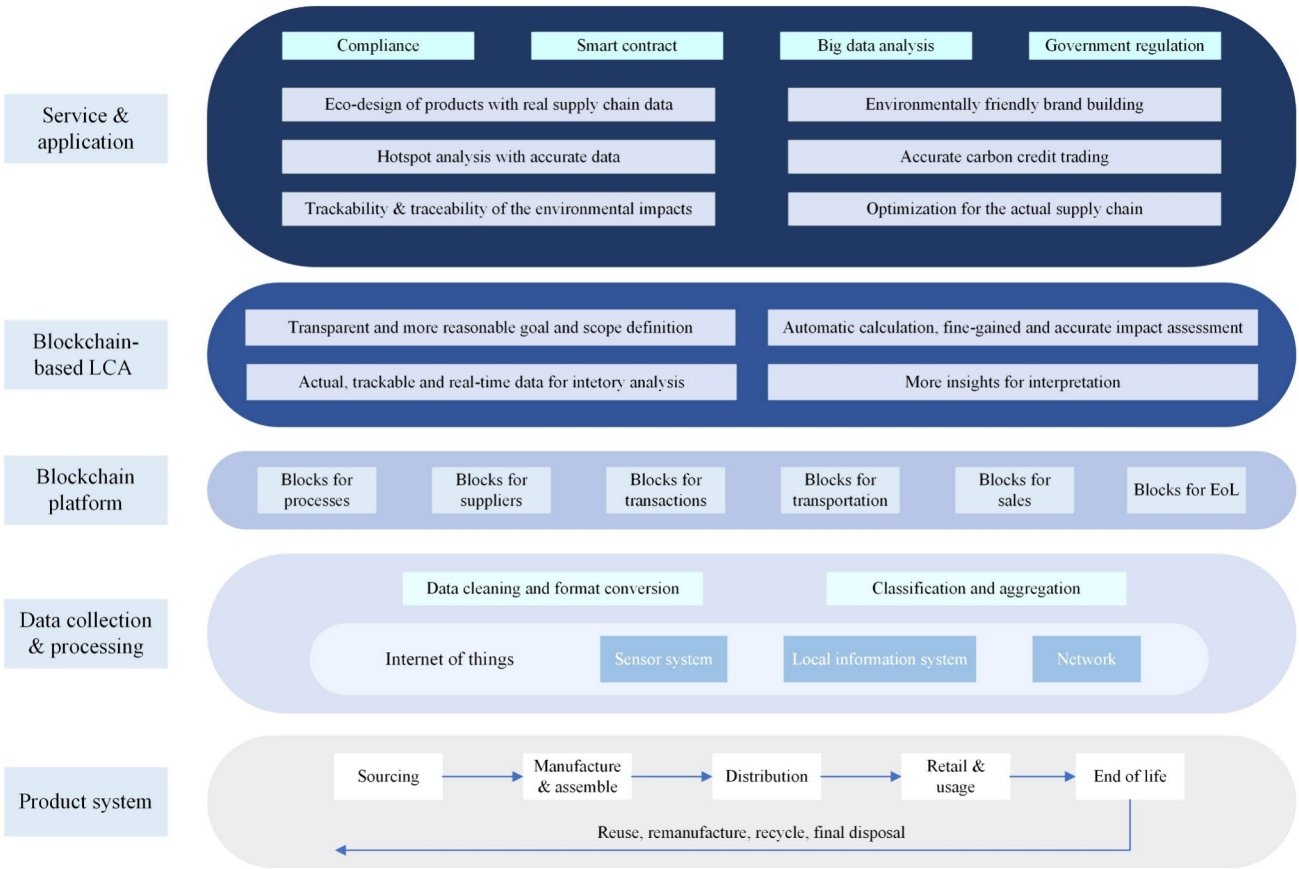


Fig. 7 New generic framework for the integration of blockchain and LCA

types of blocks for different entities: suppliers, transactions, processes, transportation, sales, and end-of-life treatment. Each supplier is allocated a unique identifier with a corresponding block to record essential supplier information such as the name, address, and registration number. Transactions between suppliers also have unique identifiers with blocks that document the participants, the time of the transaction, and the transaction details, including names of materials or semi-finished products, technical parameters, and quantities. Each manufacturing process within a supplier's operations is assigned a unique identifier with a block to record detailed information, including the manufacturing processes, the materials, consumption of water and energy, emissions, waste generation, and disposal. Similarly, transportation processes, encompassing the movement of raw materials, semi-finished products, final products, and waste, are also allocated unique identifiers with corresponding blocks for recording. Sales information, such as the location, time, and quantity, also receives unique identifiers with blocks, and the usage phase invites user participation. The end-of-life treatment stages are recorded using unique identifiers and blocks. These blocks detail the processing methods, product types, quantities, material and energy input, and emissions. Related blocks for the six entities provide a comprehensive overview of the product's entire life cycle, ensuring high levels of traceability and transparency. The blockchain-based LCA layer utilizes the collected product life cycle information to define the goals and scope according to specific requirements. Thanks to the distributed blockchain network and the traceable, transparent, and detailed inventory from it, this layer automatically calculates accurate and fine-grained life cycle environmental impact results with high efficiency, thereby providing more insights in time for the interpretation. Benefiting from the Security Hash Algorithm (SHA) of blockchain, the original data can be kept confidential, allowing only the users with keys to access it, while relevant environmental data can be selectively shared and disclosed. For example, when supplier A sends one ton of material to supplier B, the transition information will be broadcast and shared in the network. However, the detailed information such as the material, amounts, and traders is enciphered, while the environmental information is automatically calculated and open to the stakeholders or the public.

The service and application layer shows six primary applications and four potential future applications. Thanks to the inherent transparency provided by blockchain, this integration enables the full-process tracking of a product's environmental impacts. It can conduct hotspot analysis utilizing actual and real-time supply chain data, which can help to gain actual specific data in different conditions instead of average data for facilitating product ecological design and modifications. Precise carbon emission data allows for more accurate carbon credit trading. Manufacturers can share supply chain environmental impact results with consumers, thus improving their environmental brand building. The acquisition of production and environmental data throughout the supply chain serves as a solid foundation for optimizing supply chain operations. Furthermore, this integration holds potential for additional applications, such as aiding businesses in managing compliance, facilitating more transparent supply chain management for governments, integrating with smart contracts, and conducting in-depth big data analysis for further insights.

4.3.2 Comparison with other integration frameworks

This study developed a new generic blockchain-LCA integration framework. While several articles address the framework of blockchain-LCA integration, only two focus on the development of generic frameworks. Taranto et al. (2023) integrated blockchain with LCA, using LCA for the environmental evaluation and monetizing its impacts, while employing blockchain technology for transactions and records for it. Different from our comprehensive framework, their framework only focuses on the monetization and trading of environmental impacts, lacking content on data collection and processing, environmental impact calculation, and various comprehensive system applications. Similar to our work, Zhang et al. also developed a comprehensive blockchain-LCA integration framework. However, this framework is more vague regarding the relevant technical mechanisms of blockchain, with somehow fuzzy descriptions of data transmission and processing, and lacks detailed technical explanations. In contrast, our framework provides a detailed description of specific blockchain platforms, allocates different blocks to various entities and transactions, and elaborates on data collection, processing, and blockchain-related execution technical mechanisms. Moreover, our newly developed framework highlights the significant enhancement role of the blockchain-LCA system in LCA, which is fundamental for the system's various services and applications. This aspect is omitted in their framework. At the application level, we innovatively propose using the blockchain-LCA integration system for compliance supervision, establishing smart contracts for carbon trading, and enabling life cycle carbon tax management.

In summary, compared to previously developed frameworks, our comprehensive framework more thoroughly describes the functionalities of the blockchain-LCA system at different levels. It provides more detailed technical process information on applying blockchain, including how to use blockchain to collect and process data from various parts of value chains, making it more substantial and executable. Additionally, we offer advanced application scenarios such as smart contract-based carbon trading, compliance supervision, and life cycle carbon tax management.

4.3.3 Potential case study

To better illustrate the utility and effectiveness of our blockchain-LCA integration framework, we use the supply chain of cotton clothing production as an example, as shown in Fig. 8. The production of cotton clothing mainly includes five life cycle stages: cotton planting and harvesting, cotton fabric production, clothing manufacture, clothing sales and usage, and the EoL of clothing.

In our framework, the blockchain platform provides six types of blocks for different entities: suppliers, transactions, processes, transportation, sales, and end-of-life treatment. Each supplier in the cotton clothing supply chain is assigned a unique identifier and corresponding block. Transactions between suppliers involving cotton, cotton fabric, and cotton clothing also have corresponding blocks to record details such as type, batch, quantity, time, etc. Regarding the production process, IoT technology is used to collect inventory data and generate corresponding blocks. For example, in cotton clothing production, data on the consumption of energy and materials like different types of cotton fabrics, dyes, water, electricity, and emissions such as CO_2 emissions, SO_2 emissions, and heavy metal emissions are recorded, along with the recycling or disposing of solid waste like scraps. Similarly, during the transportation of products and raw materials, corresponding blocks are generated to record the information. Each batch of produced cotton clothing has a unique identifier and corresponding block, with the identifier printed on the clothing label. Retailers can share their sales information using the product's unique identifier. The inventory data gathered through the blockchain platform are refined and then processed with automated LCA software to calculate the environmental impacts for different clothing batches. Consumers can scan the unique identifiers on the clothing to access their environmental information. Moreover, when consumers dispose of old cotton clothing, they can use specially developed software to scan the identifiers and upload the disposal information to generate blocks for EoL, enhancing the further traceability of environmental impacts during the EoL phase.

Through this blockchain-LCA integration system, stakeholders can track the production information of different styles of cotton clothing for each batch and obtain accurate, trustworthy, transparent, and real-time environmental impacts. Clothing manufacturers can identify environmental hotspots based on these results, aiding in eco-design and supply chain optimization. They can also highlight the environmental attributes of their products in marketing. For transactions among suppliers, the platform supports automated transactions through blockchain-enabled smart contracts, and it can also be used for carbon trading with blockchain. Governments can use the information recorded

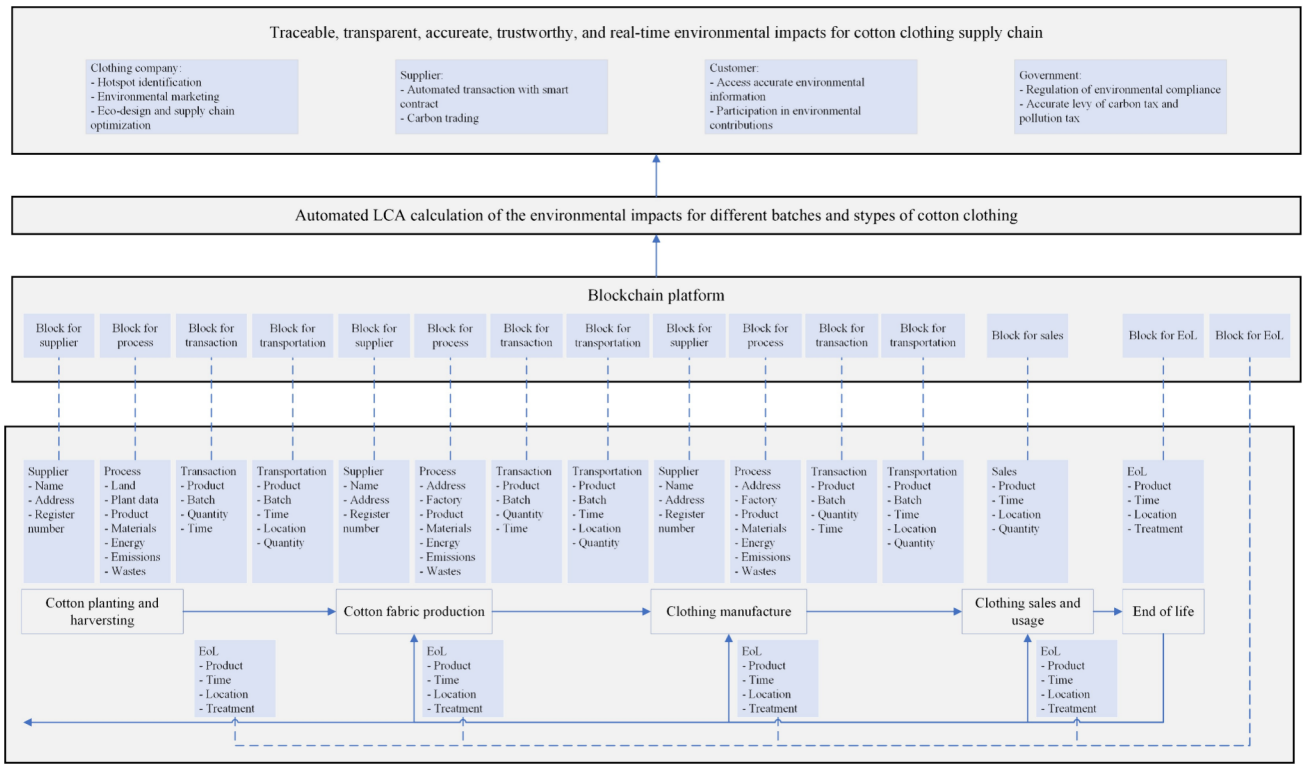


Fig. 8 Potential case study of cotton clothing supply chain

to regulate the compliance of enterprises in the supply chain and more accurately levy carbon taxes and other pollution taxes. Consumers can easily access accurate information about the environmental impacts of cotton clothes and participate in contributing to traceability and environmental information sharing during the EoL stage.

4.4 Implications

The integration of blockchain and LCA equips stakeholders of value chains with detailed, precise, and transparent information about the environmental impacts. It enables companies to identify environmental hotspots, enhance sustainable supply chain management, and optimize emission reduction efforts. Employing blockchain in the supply chain may also prevent fraudulent practices, improve the supply chain's reputation, and foster environmental responsibility among various enterprises involved.

Given the substantial benefits for businesses, this blockchain-LCA integration also creates significant commercial opportunities for technology providers, especially the integration schemes incorporating LCA data collection and calculation platforms, standard integration models, the Internet of Things, and big data analytics. Moreover, there is a growing market for developing supporting technologies and databases, such as hardware and software for identifying and quantifying environmental impacts, automated LCA, and interactive LCA databases. The complexity of blockchain-LCA solutions could necessitate technology providers to set up new departments and hire professionals to push the boundaries of current technologies.

Effective implementation of blockchain-LCA requires establishing a collaborative environment at the supply chain level, promoting information sharing through associations and alliances, such as Catena-X in the automotive industry (Schöppenthau et al. 2023). Furthermore, relevant businesses, such as governments, regulatory bodies, non-governmental organizations, industries, and communities, must enhance external stakeholders' understanding of the potential of the integration. By intensifying promotional efforts and fostering collaboration among these groups, businesses can significantly expand the market applications and influence of BC-LCA. Additionally, clearly defining roles and responsibilities within collaborations, along with careful management of data access rights and decision-making authorities, is essential to ensure successful and smooth cooperation.

The integration of blockchain and LCA also has substantial policy implications. To enhance collaboration and data sharing among supply chain businesses, governments should collaborate with businesses, industry bodies, and NGOs to develop comprehensive guidelines and standards for blockchain-LCA integration and offer financial incentives or subsidies for related projects. To prevent misuse of data transparency and ensure fairness, governments should regulate the integration process, including data permissions. Using the blockchain-LCA system, governments can manage environmental regulations more effectively. Based on accurate carbon emissions data and environmental impact assessments, the integration system can be used to regulate the enterprise's environmental compliance and enable the life cycle environmental tax initiatives as well as blockchain-based carbon trading. Additionally, establishing funds to support sustainable practices, increasing investment in relevant scientific research, and promoting professional training in blockchain and LCA are vital.

For the scientific community, fostering interdisciplinary collaboration is essential given the practical and interdisciplinary nature of this field, which includes LCA, blockchain technology, software development, and hardware deployment. It is also crucial to promote the partnerships between academia and industry to develop impactful applications. These research efforts can also enhance the LCA databases, providing lots of data that benefits researchers in related studies.

5 Conclusion

In this paper, we conducted a systematic review of the literature on the integration of blockchain and LCA. We found 31 articles from journals and conferences and extracted and analyzed the bibliographic information related to the topic. The temporal distribution, the geographical distribution, the involved industrial sectors, and the distribution among different journals and conferences are analyzed, showing that the number of studies on blockchain-LCA integration is increasing, but this field is still in the early stages. The thematic analysis revealed that the potential benefits and barriers were summarized comprehensively in current studies, but most of them lacked verification, while a limited number presented the frameworks, models, and case studies. As for the several case studies, only one paper used the real-world data collected with blockchain, while the other three used simulation data. Based on the results, we provided suggestions for the future development of the integration of blockchain and LCA, including the need for case validations, standardized guidelines, interaction platforms, encouragement policies and programs, specialized forums, and so on. In addition, we proposed a new generic systematic framework of blockchain-LCA integration to provide guidance for its implementation in the future. Moreover, we provided managerial implications, policy implications, and implications for the scientific community.

Despite the contributions mentioned above, this study also has some limitations. The division of themes in the

theme analysis is based on the author's interpretation, which could introduce a potential bias. The integration framework proposed in this paper is universal and comprehensive, but it currently lacks consideration of local economic conditions and socio-cultural factors, as well as detailed variations within different industry supply chains. Practical applications may therefore require adaptive adjustments tailored to specific contexts. Additionally, it is essential to recognize that LCA and blockchain are continuously developing, and this framework, established under current conditions, will need iterations to incorporate new technologies and methodologies. Besides, this paper only describes potential case studies within the framework, and it does not conduct actual case validations nor analyze the environmental impacts of the system. In the next step, our work will be dedicated to using the framework to conduct case studies based on real-world data and to quantitatively evaluate the environmental influences of the proposed integration system.

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Data availability Data will be made available on request.

Declarations

Competing interests The authors declare no competing interests.

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